HD 31435: a Five Per Cent Polarisation Assigned to an Unreddened Sun-like Star at 100 pc

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Abstract

Starlight from a quiet dwarf in front of the interstellar dust is polarised at parts per million, and a structure that reflects part of a star's light polarises it, so a nearby, unreddened dwarf polarised at the per cent level has no ordinary explanation at the star. We test the 28704 stars of the Panopoulou et al. compilation of optical polarimetry that have five or more neighbours within 3° and 20 per cent in distance for a Stokes vector their neighbours do not share. Four dwarfs within 200 pc are discordant at 5σ; three are a cloud's own dust, a binary in a crowded field and an interstellar sightline that Gaia and Planck recognise. The fourth, HD 31435, an F9/G0 dwarf at 100 pc, is listed at 5.48 ± 0.15 per cent and 47° in Johnson V from one measurement of 1977 November, where the eleven stars of its field within 130 pc read 0.27 per cent or less. Nothing at the star accounts for it: Gaia gives no reddening, the star is single and steady in Gaia's astrometry and velocities and in TESS, K2, Hipparcos and the Harvard plates, and WISE sees its photosphere to 22 μm. A reflector would have to return several per cent of its light, grey to 0.3 per cent from 400 to 700 nm in its Gaia XP spectrum against 40 twins; Gaia's image statistics and Pan-STARRS leave no room for it from 10 to 1300 au, and within 10 au only one that absorbs nothing and holds still escapes WISE and the light curves. The angle is the interstellar field's behind the Taurus clouds: the 40 stars of the field beyond 160 pc read a median 48°, and Planck's 353 GHz polarisation at the position predicts 49 to 54°. The amount is what background stars there reach on efficient sightlines, three of those 40 at 5.5 per cent or more. HD 283904, a reddened A2 star of the same brightness behind the cloud, lies 5′ west at the same declination to 27″, 23 s of right ascension away. The measurement is that of a star behind the cloud, most likely HD 283904 through a slipped setting, and one polarimetric measurement of the two stars decides.

1 Introduction

The light of a nearby, inactive dwarf is polarised at a few parts per million: the brightest southern stars within the Local Bubble, measured at that level, carry interstellar polarisations of tens of ppm and intrinsic ones too small to separate [1]. Interstellar polarisation grows with the dust in front of a star and is bounded by it, $p \leq 9\,E(B-V)$ per cent in the visible [2], so a star in front of the dust carries almost none. Reflection polarises: sunlight scattered from a planet's atmosphere or reflected from a flat surface is polarised by an amount set by the scattering angle and the material, and a structure that returns part of a star's light toward Earth would add that polarised light to the star's. A dwarf in front of the dust whose polarisation reaches the per cent level therefore has no explanation at the star that the archives would not show, and one that a reflector would give.

Optical polarimetry of stars has been gathered from many programmes into compilations [3, 4], and with Gaia distances each star can be set against its neighbours on the sky at the same distance, which share its interstellar column. This note reports that test and the one star it leaves, HD 31435, and reads its measurement against every archive that bears on it.

1.1 Contributions

1. A test of 28704 stars with optical polarimetry and Gaia distances against their neighbours' Stokes vectors, with the measurements reassigned among neighbours as the null.
2. For HD 31435, the one discordant dwarf left, the constraints of Gaia, WISE, TESS, K2, Hipparcos, DASCH, Gaia XP and Pan-STARRS on any source of polarised light at the star, and the reading of its measurement against the field, Planck's dust polarisation and the star of the same brightness beside it.

2 The screen

The compilation of Panopoulou et al. [4] holds 55742 measurements from 81 publications, with Gaia EDR3 identifiers and distances; 37210 stars carry Stokes q and u. Of these, 29442 are unflagged and have five or more neighbours within 3° and within 20 per cent of their distance, and 28704 of them can be judged against neighbours from their own catalogue, which removes the zero-point differences between programmes. A star's residual is its Stokes vector less the neighbours' mean, divided by its own error and the neighbours' scatter in quadrature. The same statistic with each field's measurements reassigned among its stars is the null. Residuals above 3σ occur in 8.3 per cent of the stars against 7.4 per cent in the null, and above 5σ in 1.2 against 0.9 per cent, so discordance at that level is mostly the scatter of real sightlines.

We keep dwarfs within 200 pc, where the interstellar column is small, whose residual reaches 5σ, exceeds 0.1 per cent and five times the field's scatter, and departs from the field in angle or lies in an unpolarised field. Four remain. HD 142832, an F3 dwarf at 158 pc, was measured in the survey of the Lupus 1 cloud of Alves and Franco [5] and lies at the cloud's distance, where its dust polarises it and not its nearer neighbours. Dachs SMC 2-61, a G5 dwarf at 134 pc with a renormalised unit weight error (RUWE) of 8.9 and an X-ray counterpart, was measured in a crowded field toward the Small Magellanic Cloud and is a binary or a blend. HD 127769, an A7 dwarf at 187 pc, read 1.3 per cent at 43° in the compilation of Heiles [3] against its field's 0.4 per cent at 87°; Gaia gives it an extinction $A_G = 0.32$ mag where its nearer neighbours have none, and Planck's 353 GHz dust emission at its position is polarised at 18 per cent with a field that predicts 52° and 1.9 per cent for starlight behind the full column, so its measurement sits on that sightline. The fourth is HD 31435.

3 HD 31435

3.1 The measurement

HD 31435 (Gaia DR3 154235505416809344) is an F9/G0 dwarf at 100.1 pc with $G = 8.90$, $G_{\rm BP} - G_{\rm RP} = 0.73$ and $M_G = 3.85$, at the main-sequence turnoff of an 8 to 9 Gyr population. The compilation gives $q = -0.38 \pm 0.15$ and $u = +5.47 \pm 0.15$ per cent, $p = 5.48 \pm 0.15$ per cent at position angle $47.0 \pm 0.8°$ in Johnson V, from a Minipol [6] measurement on JD 2443471.5, 1977 November 22, published by name in Table 2 of Moneti et al. [7], whose survey measured field stars toward the Taurus dark clouds. Against the mean of its neighbours, 0.04 per cent at 77°, the residual is 5.46 per cent at 46.8°, 28.9σ. No other polarimetry of the star is found among the 135 VizieR catalogues with entries at its position, and the other 93 stars of that programme read below 0.4 per cent within 130 pc and rise with reddening beyond it.

Figure 1 shows the field. The compilation holds 58 stars within 3° of HD 31435: the 11 within 130 pc read 0.27 per cent or less, and the 40 beyond 160 pc, behind the Taurus clouds, read a

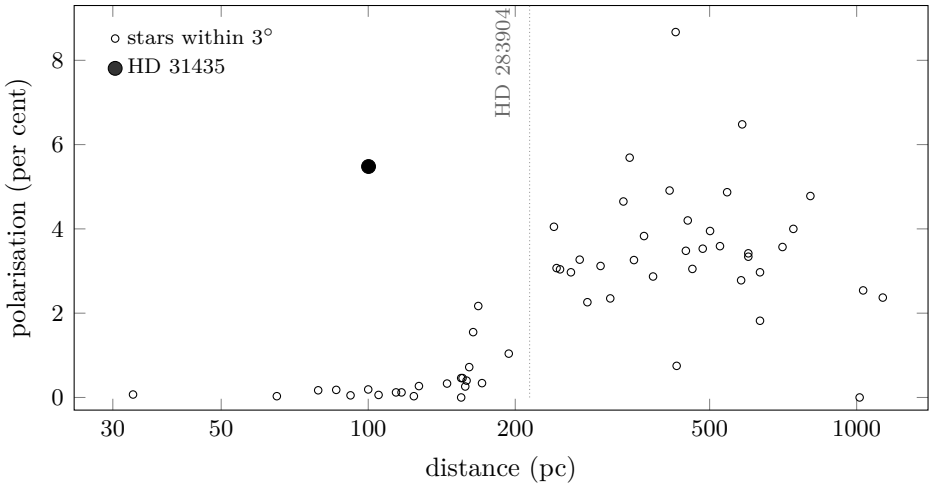


Figure 1: Optical polarisation against Gaia distance for the 58 stars of the compilation within 3° of HD 31435, with HD 31435 filled. The dotted line marks the distance of HD 283904, which has no measurement.

median of 3.2 per cent, with a 90th percentile of 4.9 per cent and three of the 40 at 5.5 per cent or more, at a median angle of 48° and a 16th to 84th percentile range of 35 to 61°. HD 31435 lies at the distance of the unpolarised stars with the polarisation of the background ones.

3.2 The star

Nothing at the star gives per cent polarisation. Interstellar polarisation of 5.48 per cent needs $E(B-V) \geq 0.6$ mag [2]; Gaia's astrophysical parameters give $E(G_{\rm BP}-G_{\rm RP}) = 0.00$, and the star's $B-V$ of 0.565 lies 0.015 mag blueward of the intrinsic colour of its type. The WISE photometry [8] follows the photosphere from 3.4 to 22 μm, with no warm dust. Its astrometry carries no companion: RUWE 0.99, no multiple image peaks, and a Hipparcos–Gaia proper-motion anomaly [9] of $20 \pm 32\,{\rm m\,s^{-1}}$. Gaia's radial velocity, from 17 transits over 920 days, varies by less than $1\,{\rm km\,s^{-1}}$. TESS [10], K2 [11] and Hipparcos [12] photometry is flat to 0.03 per cent, and the Harvard plates [13] show no dimming around 1977 against matched stars on the same plates. The star is an old, single, quiet dwarf in front of the dust, whose intrinsic polarisation would be of order ten ppm.

3.3 A reflector

Light reflected by a structure near the star and seen together with it is polarised by $p \approx f P_r$, with f the fraction of the star's light returned toward Earth and P_r the polarisation of the reflected light, so 5.48 per cent needs $f \geq 0.055$ even if every reflected photon is polarised. Specular reflection from metals polarises weakly away from grazing incidence: by the Fresnel relations aluminium gives 3 to 14 per cent at incidence angles of 45 to 80° and silver less than 2 per cent, so a metal mirror needs f of order one, and a dielectric reaches 98 per cent at Brewster's angle while reflecting 9 per cent of what falls on it, so it must intercept more than half of the star's light. Only a reflector that sends back one polarisation and passes or absorbs the other, as a conducting grid does, gives 5.5 per cent with $f \approx 0.06$.

The reflected light carries the reflector's colour. Against the median of 40 Gaia XP spectra [14] of unreddened twins matched in temperature, gravity and metallicity, HD 31435's spectrum agrees to 0.3 per cent from 400 to 700 nm (0.9998 ± 0.0013, 0.9970 ± 0.0005 and 1.0013 ± 0.0011 in the

three 100 nm bands) and lies 0.35 and 0.66 per cent high at 700 to 800 and 800 to 900 nm, within the twins' own spread of 0.7 to 1.0 per cent there; its slope, +0.0014 per 100 nm, lies inside the twins' −0.005 to +0.006. Aluminium's reflectance dips by about six per cent near 830 nm, so a tenth of the light reflected from aluminium would take 0.6 per cent out of the 800 to 900 nm band, where HD 31435 lies above its twins rather than below them. The spectrum is that of a star with nothing coloured added to it, and leaves a grey reflector.

The reflector must also lie where no archive sees it. A structure within about 10 au that absorbs as little as a per cent of what it intercepts would radiate above the photosphere at 22 μm, which WISE excludes; one that absorbs nothing is not excluded there, and TESS, K2 and Hipparcos, flat to 0.03 per cent over years, exclude one whose reflected light changes along an orbit, so what is left inside 10 au is a perfect reflector that holds still. Gaia's image parameters, with RUWE 0.99 and no second peak, exclude a resolved source of a few per cent of the star's light at 0.1 to 2″, 10 to 200 au. Pan-STARRS [15] images in g, r, i, z and y, with each radial profile normalised at 13 to 25″ against three stars of like brightness in the same frames, show no excess light at 2 to 13″ in any filter, so no extended reflector lies at 200 to 1300 au.

3.4 The sightline

The measured angle is the field's. Planck's 353 GHz maps [16], read at HD 31435's position on the HEALPix grid [17], give dust emission polarised at 6 to 8 per cent with a polarised intensity of 0.12 to 0.15 MJy sr^{-1} within 5 to 20′; turned through 90° and into the equatorial frame, the field predicts 49 to 54° for starlight behind the column, against the measured 47° and the background stars' median of 48°. The ratio of polarised submillimetre intensity to optical polarisation, 5.4 MJy sr^{-1} [18], predicts 2.3 to 2.8 per cent for a star behind the whole column, half the measured value; sightlines scatter about that ratio, and three of the field's 40 background stars reach 5.5 per cent or more.

The star of the same brightness beside HD 31435 is behind the cloud. HD 283904, an A2 star at 214 pc with $A_G = 1.3$ mag and the 1.527 μm diffuse interstellar band [19] in its APOGEE spectrum [20], lies 5′ west of HD 31435, at the same declination to 27″ and 23 s of right ascension away, and has never been measured. A setting that slipped by 23 s of hour angle would have placed it in the aperture in place of HD 31435, and its polarisation would be the field's: the angle exactly, and the amount on an efficient sightline.

4 Related work

Polarimetry of nearby stars at parts per million [1] measures the interstellar polarisation inside the Local Bubble and bounds the intrinsic polarisation of dwarfs, and the compilations of optical polarimetry [3, 4], joined to Gaia distances, are used to map the interstellar field and the dust in three dimensions. Planck's comparison of submillimetre and optical polarisation [18] gives the ratio used here to predict a star's polarisation from the dust emission at its position. We find no earlier use of these compilations to look for stars whose polarisation their neighbours do not share, nor any discussion of HD 31435's measurement.

5 Conclusion

Among 28704 stars with optical polarimetry, one dwarf within 200 pc carries a polarisation that neither its neighbours nor anything at the star accounts for: HD 31435, listed at 5.48 per cent from

a single measurement of 1977. A reflector that made it would have to return at least six per cent of the star's light, be grey across the visible to a third of a per cent and polarise like a conducting grid, either lie within 10 au, absorb nothing and hold still or lie beyond 1300 au, and be aligned by chance with the Galaxy's field behind the Taurus clouds. The measurement has instead every property of a star behind those clouds, and a star of the same brightness behind them sits 23 s of right ascension away. The measurement most likely belongs to HD 283904.

Measuring both stars in one night with any polarimeter of 0.1 per cent precision decides between the two readings, and measuring HD 31435 at parts per million would bound what a reflector could hide at the level the Local Bubble's stars set.

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